

DELIVERABLE 1: Complete Requirements Specification Table

Problem Statement #55: Book Club Reading Challenge & Discussion Portal | **Student:** Sushruth S (PES1UG24CS486)

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (Justification)	Comments / Technical Notes
FR-001	Functional	The system shall hide content inside spoiler-tagged discussion comments until the user explicitly clicks to reveal or marks that chapter as read.	High	Pass: Spoiler text masked with blur filter. Fail: Spoiler content exposed in search preview.	Protects readers from inadvertent plot spoilers before finishing chapters.	Implemented via client-side CSS blur mask with persistent user read-state sync.
FR-002	Functional	The system shall track individual user reading progress by logging daily/monthly page count goals and updating progress percentage toward the active challenge.	High	Pass: Progress bar increments immediately upon logging valid integer page count. Fail: Negative values accepted or gauge fails to update.	Core gamification mechanic to motivate members to complete scheduled reading milestones.	Input validated against total book page bound; updates user dashboard asynchronously.
FR-003	Functional	The system shall allow Discussion Leads to create and moderate chapter-specific discussion threads with custom pinned discussion prompts.	High	Pass: Only verified Discussion Leads can pin/close threads or delete flagged comments. Fail: Regular members access moderation controls.	Facilitates structured, organized, and toxicity-free book analysis across chapters.	Enforced using Role-Based Access Control (RBAC) on backend moderation API endpoints.
FR-004	Functional	The system shall allow Club Members to cast exactly one vote in the active monthly book selection poll from a curated list of nominated titles.	High	Pass: Member submits 1 vote, receives confirmation, voting controls disable. Fail: Member submits multiple votes upon refresh.	Ensures democratic, fair, and tamper-resistant book selection for upcoming reading cycles.	Guaranteed by unique compound index (UserID + PollID) in persistence layer.
FR-005	Functional	The system shall award and display digital achievement badges on user profiles upon reaching predefined reading milestones (e.g., 500 pages logged).	Medium	Pass: Badge icon renders in user profile showcase within 5s of milestone trigger. Fail: Badge granted without reaching page goal.	Increases long-term member engagement and social recognition within the club.	Triggered via background event listener on goal progress update.
NFR-001	Non-Functional (Perf & Sec)	Monthly poll voting results must prevent duplicate votes using user session verification and update tallies in real-time.	High	Pass: Benchmarking confirms latency < 200ms and 0 duplicate votes under 1,000 peak users. Fail: Duplicate vote accepted or lag > 1.5s.	Ensures poll data integrity and instantaneous community feedback during high-traffic voting.	Utilizes in-memory Redis tally cache paired with transactional DB writes.
NFR-002	Non-Functional (Security & Availability)	The portal shall maintain 99.9% system availability and encrypt all stored user reading history and credentials using AES-256 and SHA-256.	High	Pass: Monthly uptime logs ≥ 99.9% and automated vulnerability scans confirm zero plaintext PII leaks. Fail: Downtime > 43.8 min/month.	Protects member privacy, avoids credential compromise, and ensures reliable access during reading deadlines.	Enforces TLS 1.3 in transit and column-level encryption at rest with automated health probes.